Physics 9702 Paper 4 · Topic 23

Nuclear physics

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/41 May/June 2022, Question 8 (12 marks) page 2

Examiner report: (ii) Candidates found it difficult to recall and draw the correct line. (ii) Candidates found this explanation challenging.

2 9702/42 October/November 2020, Question 12 (7 marks) page 4

Examiner report: (b) Candidates found this question difficult, and a large proportion of candidates demonstrated a fundamental confusion between the concepts of nuclide mass and nucleon number. Those candidates who started by considering the variation with nucleon number of binding energy per nucleon were usually able to achieve partial credit, but only the very strongest candidates were able to give a complete answer and achieve full credit.

3 9702/41 May/June 2023, Question 9 (9 marks) page 5

Examiner report: (c) (i) Very few candidates realised the key point here that the probability of decay of a nucleus per unit time is constant, which is why the decay is random.

4 9702/42 May/June 2023, Question 9 (11 marks) page 6

Examiner report: (a) The definition of mass defect appears to be not as well known as most definitions, and many candidates did not give a correct answer. (c) (i) This was a challenging question.

5 9702/42 October/November 2021, Question 12 (8 marks) page 8

Examiner report: (b) (i) Candidates found this to be a challenging question. Candidates that chose to calculate the activities often struggled with the fact that the data was presented as logarithms of activity, and those that used the gradient route (to find decay constant and hence half-life) often had difficulty because the y-axis did not start at the origin.

6 9702/41 October/November 2022, Question 10 (10 marks) page 10

Examiner report: (iv) Many candidates were unable to take natural logarithms of the exponential decay equation and show that the gradient of the line is $-\lambda$. (c) This explanation proved to be challenging.

7 9702/42 February/March 2024, Question 8 (9 marks) page 12

Examiner report: (iii) This was a challenging question, and many candidates gave responses that were along the right lines but not enough to be awarded credit. A common misconception was that the lack of stability was due to there being more neutrons than protons, overlooking the fact that this is true for the vast majority of stable nuclei (including xenon-132).

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

8 **(a)** **(i)** State what is meant by nuclear binding energy.

.....

.....

..... [2]

(ii) On Fig. 8.1, sketch a line to show the variation with nucleon number A of the binding energy per nucleon E of a nucleus.

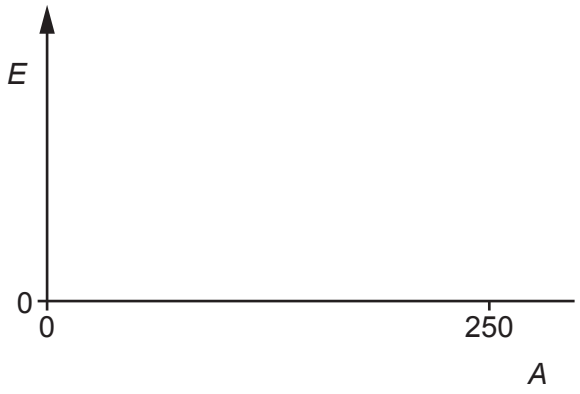


Fig. 8.1

[2]

(b) In one type of nuclear process, deuterium ($^{2}_{1}H$) undergoes the reaction

$$^{2}_{1}H + {}^{2}_{1}H \longrightarrow {}^{3}_{2}He + {}^{1}_{0}n.$$

(i) State the name of this type of nuclear process.

..... [1]

(ii) Explain, with reference to your line in **(a)(ii)**, why this reaction results in the release of energy.

.....

.....

..... [2]

(c) Table 8.1 shows the masses of the particles involved in the reaction in **(b)**.

Table 8.1

particle	mass / u
$^{1}_{0}n$	1.008 665
$^{2}_{1}H$	2.014 102
$^{3}_{2}He$	3.016 029

Calculate the energy released when 1.00 mol of deuterium undergoes the reaction.

energy = ... J [5]

[Total: 12]

12 (a) (i) Define nuclear *binding energy*.

.....

.....

..... [2]

(ii) Explain what is meant by a *nuclear fission* reaction.

.....

.....

..... [2]

(b) A student suggests that one possible nuclear reaction is

$$^{56}_{26}\mathrm{Fe} + ^{1}_{0}\mathrm{n} \longrightarrow ^{20}_{9}\mathrm{F} + ^{37}_{17}\mathrm{C}l.$$

The binding energy per nucleon of a nucleus varies with the nucleon number.
Use this variation to explain why the reaction would **not** result in an overall release of energy.

.....

.....

.....

.....

..... [3]

[Total: 7]

9 Carbon-11 is radioactive and decays by β^+ emission to form boron-11. Carbon-11 has a half-life of 20 minutes. Boron-11 is stable.

(a) Define half-life.

.....

..... [1]

(b) A sample contains N_0 nuclei of carbon-11 and no other nuclei at time $t = 0$.

On Fig. 9.1, sketch the variation with t of the number of nuclei of **boron-11** in the sample.

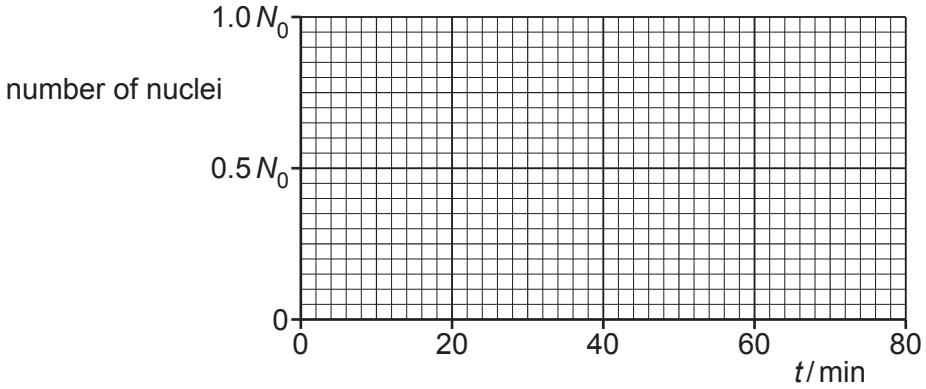


Fig. 9.1

[3]

(c) **(i)** Explain, with reference to the random nature of radioactive decay, why the activity of the carbon-11 sample in **(b)** decreases with time.

.....

.....

..... [2]

(ii) State, with reasons, whether a radiation detector placed near to the sample of carbon-11 indicates a measured count rate from the sample that is less than, the same as or greater than the activity of the sample.

.....

.....

.....

.....

..... [3]

[Total: 9]

9 **(a)** Define mass defect.

.....

.....

..... [2]

(b) Table 9.1 shows the mass defects of three nuclei.

Table 9.1

nucleus	mass defect / u
$^{2}_{1}H$	0.002 388
$^{3}_{1}H$	0.009 105
$^{4}_{2}He$	0.030 377

The nuclear fusion process in a particular star is described by

$$^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + X$$

where X is a particle that has no mass defect.

(i) State the name of particle X.

..... [1]

(ii) Show that the energy released when one nucleus of $^{4}_{2}He$ is formed in this fusion reaction is 2.8×10^{-12} J.

[3]

(c) The star in **(b)** has a radius of 2.3×10^9 m and a luminosity of 1.4×10^{28} W.
All the energy released from the formation of $^4_2$He is radiated away from the star.
All the energy that is radiated from the star has been released in the formation of $^4_2$He.

Determine:

(i) the mass of $^4_2$He produced per unit time by the fusion process

mass per unit time = ... $kg\,s^{-1}$ [3]

(ii) the surface temperature of the star.

temperature = ... K [2]

[Total: 11]

12 **(a)** Radioactive decay is both random and spontaneous.

State what is meant by:

(i) *random*

.....

..... [1]

(ii) *spontaneous*.

.....

..... [1]

(b) A sample of radioactive material contains atoms of an unstable nuclide X. The activity of the sample due to the atoms of X is *A*. The variation with time *t* of ln *A* is shown in Fig. 12.1.

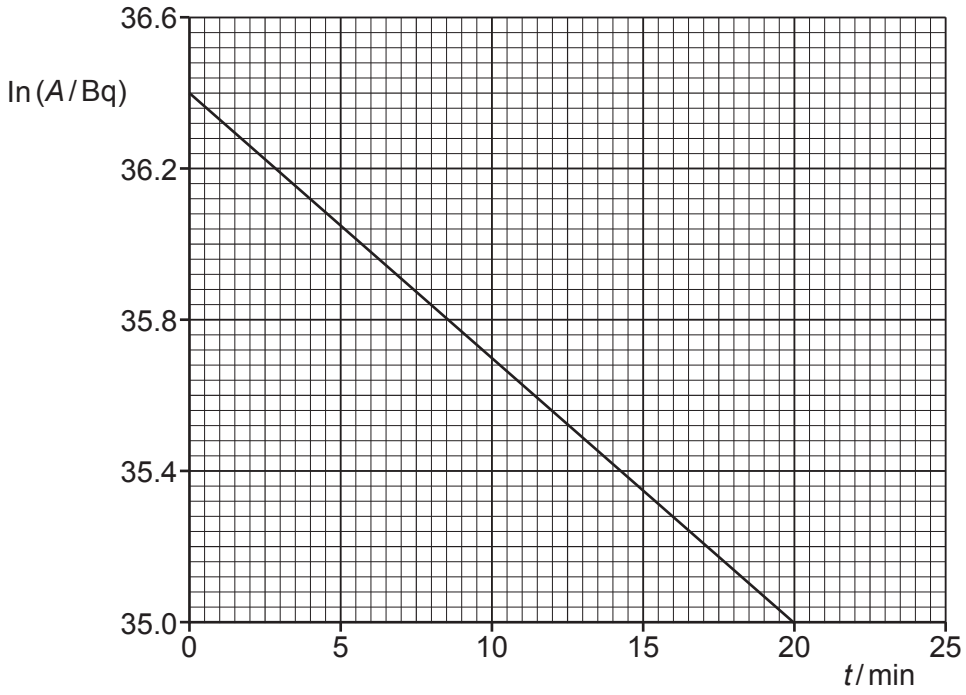


Fig. 12.1

(i) Use Fig. 12.1 to determine the half-life, in minutes, of nuclide X.

half-life = ... min [3]

(ii) At time $t = 0$, the mass of the atoms of X in the sample is 5.66×10^{-7} kg.

Determine the nucleon number of X.

nucleon number = ... [3]

[Total: 8]

10 Carbon-15 ($^{15}_{6}C$) is an isotope of carbon that undergoes radioactive decay to nitrogen-15 ($^{15}_{7}N$), which is a stable isotope of nitrogen.

Radioactive decay is both a random and a spontaneous process.

(a) State what is meant by:

(i) random

.....

..... [1]

(ii) spontaneous.

.....

..... [1]

(b) A small sample of carbon-15 decays. The mass M of carbon-15 in the sample decreases with time t.

Fig. 10.1 shows the variation with t of the value of ln ($M / 10^{-16}$ g).

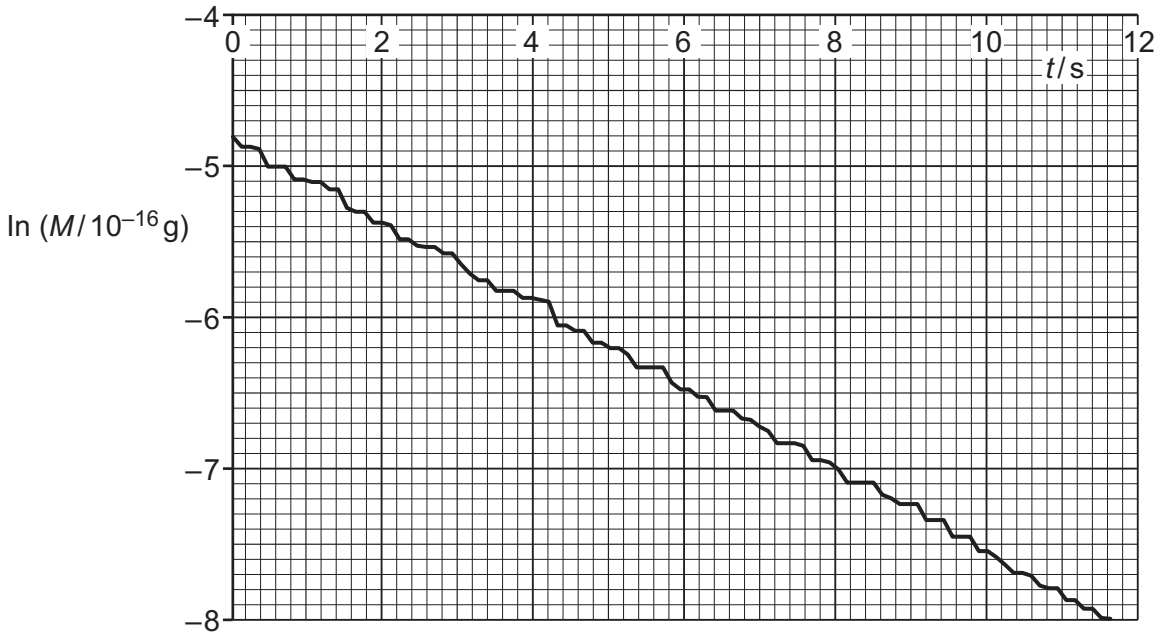


Fig. 10.1

(i) State how Fig. 10.1 demonstrates that radioactive decay is random.

.....

..... [1]

(ii) On Fig. 10.1, draw the straight line of best fit. [1]

(iii) Show that the decay constant λ of carbon-15 is given by the magnitude of the gradient of your line in **(b)(ii)**.

[1]

(iv) Use your line in **(b)(ii)** to determine λ. Give a unit with your answer.

λ = ... unit ..... [2]

(v) Use your answer in **(b)(iv)** to calculate the half-life of carbon-15.

half-life = ... s [1]

(c) The equation for the decay of carbon-15 can be written as

$$^{15}_{6}\mathrm{C} \longrightarrow {}^{15}_{7}\mathrm{N} + {}^{0}_{-1}\beta + {}^{0}_{0}\bar{\nu}.$$

State and explain how the mass of the products of the decay must compare with the mass of the carbon-15 nucleus.

.....

.....

..... [2]

[Total: 10]

8 **(a)** State what is meant by the binding energy of a nucleus.

.....

.....

..... [2]

(b) A nucleus of uranium-235 absorbs a neutron and becomes unstable. It then undergoes a fission reaction. One possible reaction is

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{142}_{54}\text{Xe} + ^{90}_{38}\text{Sr} + \text{neutrons}.$$

(i) Determine the number of neutrons produced in this fission reaction.

number = ... [1]

(ii) Data for the binding energies per nucleon for this fission reaction are given in Table 8.1.

Table 8.1

isotope	binding energy per nucleon / MeV
uranium-235	7.59
xenon-142	8.37
strontium-90	8.72

Calculate the energy released, in MeV, from the fission of one nucleus of uranium-235.

energy = ... MeV [2]

(iii) The isotope xenon-142 is unstable. The isotope xenon-132 is stable.

Suggest a reason why xenon-142 is unstable.

.....

..... [1]

(iv) Xenon-142 decays into the isotope caesium-142.

A sample initially contains only nuclei of xenon-142. After a time equal to 6.0 s, the ratio

$$\frac{\text{number of decayed nuclei of xenon-142}}{\text{number of undecayed nuclei of xenon-142}}$$

is equal to 31.

Calculate the half-life of xenon-142. Show your working.

half-life = ... s [3]

[Total: 9]

Question	Answer	Marks
8(a)(i)	energy required to separate the nucleons (in the nucleus)	**M1**
	to infinity	**A1**
8(a)(ii)	curve starting close to the origin and forming a single peak	**B1**
	peak shown to left of centre, with steep line on LHS of peak and shallow line on RHS of peak	**B1**
8(b)(i)	fusion	**B1**
8(b)(ii)	both particles have low A values **or** both particles are at left-hand end of graph	**B1**
	He-3 has higher binding energy (per nucleon) than H-2	**B1**
8(c)	$\Delta m = [(2 \times 2.014102) - (3.016029 + 1.008665)]$ u $(= 0.00351$ u$)$	**C1**
	$E = \Delta mc^2$	**C1**
	$= 0.00351 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ $(= 5.24 \times 10^{-13}$ J$)$	**C1**
	1.00 mol of deuterium forms 0.500 mol of helium-3	**C1**
	total energy $= 0.500 \times 6.02 \times 10^{23} \times 5.24 \times 10^{-13}$ $= 1.58 \times 10^{11}$ J	**A1**

Question	Answer	Marks
12(a)(i)	energy required to separate nucleons (of nucleus)	**M1**
	to infinity	**A1**
12(a)(ii)	a (single) large nucleus divides to form (smaller) nuclei	**B1**
	any one point from: • initiated by neutron bombardment • resulting nuclei are of similar size • binding energy per nucleon increases • total binding energy increases • neutrons released • combined mass of smaller nuclei is less than mass of large nucleus	**B1**
12(b)	binding energy per nucleon is a maximum at around $A = 56$	**B1**
	products of splitting a ^{56}Fe nucleus must have a lower total binding energy	**B1**
	(reaction would require) a net input of energy	**B1**

Question	Answer	Marks
9(a)	time for activity (of sample) to halve	**B1**
9(b)	sketch: line with positive gradient starting at (0,0) and extending to t = 80 min	**B1**
	exponential curve, extending from t = 0 to t = 80 min, with gradient of steadily decreasing magnitude	**B1**
	line passing through (0,0), (20, 0.5N_0) and (40, 0.75 N_0)	**B1**
9(c)(i)	every (undecayed) nucleus has the same probability of decay	**M1**
	fewer (undecayed) nuclei remaining (with time), so fewer will decay (in a given time interval)	**A1**
9(c)(ii)	• sample emits in all directions but detector only captures emissions in one direction • some emissions are absorbed before reaching detector • some emissions are scattered within the sample • simultaneous arrival of multiple particles only registers once • some particles may reach detector but not cause ionisation *Any two points, 1 mark each*	**B2**
	measured count rate is less than the activity	**B1**

Question	Answer	Marks
9(a)	difference between mass of nucleus and (total) mass of nucleons	**M1**
	when infinitely separated	**A1**
9(b)(i)	neutron	**B1**
9(b)(ii)	$E = \Delta m\, c^2$	**C1**
	Δm = (0.030377 – 0.002388 – 0.009105)u (= 0.018884u)	**C1**
	energy release = $(0.030377 - 0.002388 - 0.009105) \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 = 2.8 \times 10^{-12}$ J	**A1**
9(c)(i)	number of atoms per unit time = $(1.4 \times 10^{28}) / (2.8 \times 10^{-12})$ (= $5.0 \times 10^{39}\ s^{-1}$)	**C1**
	mass of one atom = $4 \times 1.66 \times 10^{-27}$ or $(4 \times 10^{-3}) / (6.02 \times 10^{23})$ (= 6.64×10^{-27} kg)	**C1**
	mass per unit time = $6.64 \times 10^{-27} \times 5.0 \times 10^{39}$ = 3.3×10^{13} kg s^{-1}	**A1**
9(c)(ii)	$L = 4\pi\sigma r^2 T^4$ $1.4 \times 10^{28} = 4\pi \times 5.67 \times 10^{-8} \times (2.3 \times 10^9)^2 \times T^4$	**C1**
	T = 7800 K	**A1**

Question	Answer	Marks
12(a)(i)	cannot predict when a particular nucleus will decay **or** cannot predict which nucleus will decay next	**B1**
12(a)(ii)	(decay is) not affected by external (environmental) factors	**B1**
12(b)(i)	$A = A_0 \exp(-\lambda t)$ and so $\ln A = \ln A_0 - \lambda t$ gradient of line = $(-)\lambda$	**C1**
	λ = (36.4 – 35.0) / (20 – 0) (= 0.07(0) min^{-1})	**C1**
	half-life = ln 2 / λ = ln 2 / 0.070 = 10 min	**A1**
	or	
	A_0 = exp (–36.4) = 6.43×10^{15} (Bq)	**(C1)**
	$A_0 / 2 = 3.21 \times 10^{15}$ (Bq), so ln (A_0 / 2) = 35.7	**(C1)**
	read off half-life = 10 min	**(A1)**
	or	
	(at one half-life,) ln A = 36.4 – ln 2	**(C1)**
	= 35.7	**(C1)**
	read off half-life = 10 min	**(A1)**

Question	Answer	Marks
12(b)(ii)	$A = \lambda N$	**C1**
	N = mass / (nucleon number $\times$ u) **or** N = (mass / nucleon number) $\times$ N_A	**C1**
	$\exp(36.4) = (1.17 \times 10^{-3} \times 5.66 \times 10^{-7}) / (\text{nucleon number} \times 1.66 \times 10^{-27})$ **or** $\exp(36.4) = (1.17 \times 10^{-3} \times 5.66 \times 10^{-4} \times 6.02 \times 10^{23}) / \text{nucleon number}$ nucleon number = 62	**A1**

Question	Answer	Marks
10(a)(i)	cannot predict when a (particular) nucleus will decay **or** cannot predict which nucleus will decay next	**B1**
10(a)(ii)	not affected by external / environmental factors	**B1**
10(b)(i)	line fluctuates **or** trend is a straight line	**B1**
10(b)(ii)	straight line of best fit drawn on Fig. 10.1	**B1**
10(b)(iii)	$M = M_0 \exp(-\lambda t)$ so $\ln M = \ln M_0 - \lambda t$ so gradient = $-\lambda$ (and magnitude of gradient = λ)	**B1**
10(b)(iv)	gradient = (–) (8.0 – 4.8) / (11.6 – 0) (*allow any correct pair of values from Fig. 10.1*)	**C1**
	$\lambda = 0.28\ s^{-1}$	**A1**
10(b)(v)	half-life $= 0.693 / \lambda$ $= 0.693 / 0.28$ $= 2.5$ s	**A1**
10(c)	(for reaction to occur,) energy is released	**B1**
	energy release comes from fall in mass so total mass of products must be less (than mass of carbon-15)	**B1**

Question	Answer	Marks
8(a)	(minimum) energy required to separate the nucleons (of a nucleus)	**M1**
	to infinity	**A1**
8(b)(i)	4	**A1**

Question	Answer	Marks
8(b)(ii)	energy= (142 × 8.37) + (90 × 8.72) – (235 × 7.59)	**C1**
	= 190 MeV	**A1**
8(b)(iii)	**either** it has too many neutrons (for the number of protons) **or** its neutron to proton ratio is too high	**B1**
8(b)(iv)	(when t = 6.0 s), $N / N_0 = 1 / 32$	**C1**
	either $(1 / 32) = \exp(-\ln 2 \times 6.0 / t_{½})$	**C1**
	$t_{½}$ = 1.2 s	**A1**
	or $32 / 2^n = 1$ so n = 5 (half-lives)	**(C1)**
	$t_{1/2}$ = 6.0 / 5 = 1.2 s	**(A1)**